Discrete Math W5

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ETH Zurich

Hello

I am Diego

3rd Semester CS Student

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Group Chat



Plan

1. Feedback
2. Theory Recap
3. Exercises
4. Questions?

Feedback

Theory Recap

Where Are We?

- Basic Logic & Proofs
- ⇒ Sets & Relations
- Number Theory
- Algebra
- Logic

3. Sets & Relations

- Introduction
- Sets and Operations on Sets
- Relations
- Equivalence Relations
- Partial Order Relations
- Functions
- Countable and Uncountable Sets

Relations

Definition 3.9

A *(binary) relation* ρ from a set A to a set B (also called an (A, B)-relation) is a subset of $A \times B$. If $A = B$, then ρ is called a relation *on* A.

Definition 3.10

For any set A, the *identity relation* on A, denoted id_A (or simply id), is the relation

$$\mathrm{id}_A = \{(a, a) \mid a \in A\}.$$

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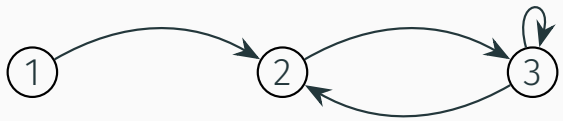
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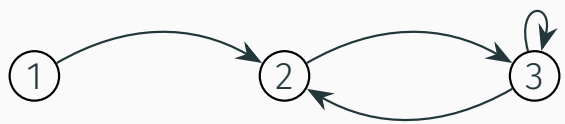
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Relations are sets, so all set operations apply ($\rho \cup \sigma$, $\rho \cap \sigma$, $\rho \subseteq A \times B$). On a finite set A there are $2^{|A|^2}$ relations on A. ($2^{|A| \cdot |B|}$ for A, B)

Inverse and Composition

Definition 3.11

The *inverse* of a relation ρ from A to B is the relation $\widehat{\rho}$ from B to A defined by

$$\widehat{\rho} \stackrel{\text{def}}{=} \{(b, a) \mid (a, b) \in \rho\}.$$

Definition 3.12

Let ρ be a relation from A to B and let σ be a relation from B to C. Then the *composition* of ρ and σ, denoted $\rho \circ \sigma$ (or also $\rho\sigma$), is the relation from A to C defined by

$$\rho \circ \sigma \stackrel{\text{def}}{=} \{(a, c) \mid \exists b\, ((a, b) \in \rho \wedge (b, c) \in \sigma)\}.$$

The n-fold composition of a relation ρ on a set A with itself is denoted ρ^n.

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Example: $\rho = \{(1, 2), (2, 3)\}$, $\sigma = \{(2, 3), (3, 1)\}$ on $\{1, 2, 3\}$:

$$\rho \circ \sigma = \{(1, 3), (2, 1)\} \qquad \sigma \circ \rho = \{(3, 2)\}$$

so composition of relations is *not commutative*.

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so composition of relations is *not commutative*.

"is child of" $\circ$ "is brother of" $=$ "is nephew of".

Socks and Shoes

Lemma 3.7

The composition of relations is *associative*, i.e., we have $\rho \circ (\sigma \circ \phi) = (\rho \circ \sigma) \circ \phi$.

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Let ρ be a relation from A to B and let σ be a relation from B to C. Then the inverse $\widehat{\rho\sigma}$ of $\rho\sigma$ is the relation $\widehat{\sigma}\,\widehat{\rho}$.

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In the morning you put on *socks* (ρ), then *shoes* (σ) $\implies$ $\rho\sigma$.

In the evening you take them off in the *reverse order*: shoes first, then socks $\implies$ $\widehat{\sigma}\,\widehat{\rho}$.

Reflexivity and Irreflexivity

Definition 3.13

A relation ρ on a set A is called *reflexive* if

$$a \ \rho \ a$$

is true for all $a \in A$, i.e., if

$$\mathrm{id} \subseteq \rho.$$

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A relation ρ on a set A is called *irreflexive* if $a \not\rho a$ for all $a \in A$, i.e., if $\rho \cap \mathrm{id} = \varnothing$.

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Examples on $\{a, b, c\}$:

$\{(a, a), (a, b)\}$ is *neither*: (b, b) is missing (not reflexive), (a, a) is present (not irreflexive).

$\varnothing$ is irreflexive but not reflexive.

Symmetry and Antisymmetry

Definition 3.15

A relation ρ on a set A is called *symmetric* if

$$a \ \rho \ b \iff b \ \rho \ a$$

is true for all $a, b \in A$, i.e., if $\rho = \widehat{\rho}$.

Definition 3.16

A relation ρ on a set A is called *antisymmetric* if

$$a \ \rho \ b \wedge b \ \rho \ a \implies a = b$$

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Examples on $\{1, 2\}$:

- $\rho = \{(1, 2), (2, 1)\}$: symmetric, not antisymmetric
- $\sigma = \{(1, 2)\}$: antisymmetric, not symmetric
- $\{(1, 1)\}$: symmetric *and* antisymmetric

Transitivity

Definition 3.17

A relation ρ on a set A is called *transitive* if

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is true for all $a, b, c \in A$.

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A relation ρ is transitive if and only if $\rho^2 \subseteq \rho$.

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Example: $\rho = \varnothing$ and $\rho = \mathrm{id}$ are (vacuously) transitive.

Transitive Closure

Definition 3.18

The *transitive closure* of a relation ρ on a set A, denoted ρ^*, is

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Example: $\rho = \{(1,2),(2,3),(3,4)\}$ on $\{1,2,3,4\}$:

$$\rho^2 = \{(1,3),(2,4)\}, \quad \rho^3 = \{(1,4)\}, \quad \rho^4 = \varnothing$$

$$\rho^* = \{(1,2),(2,3),(3,4),\\ (1,3),(2,4),(1,4)\}$$

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ρ^* relates two elements iff there is a *path* from one to the other. It is the smallest transitive relation containing ρ.

A relation is **reflexive** if for each point x ...



...there is a loop at x:



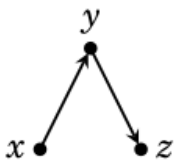
A relation is **symmetric** if whenever there is an arrow from x to y ...



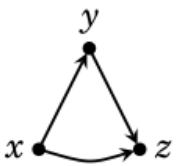
...there is also an arrow from y back to x:



A relation is **transitive** if whenever there are arrows from x to y and y to z ...



...there is also an arrow from x to z:



(If $x = z$, this means that if there are arrows from x to y and from y to x ...

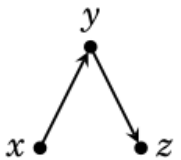
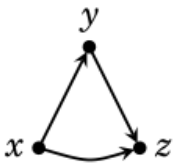


...there is also a loop from x back to x.)



A relation is **reflexive** if for each point x ...  ...there is a loop at x: 

A relation is **symmetric** if whenever there is an arrow from x to y ...  ...there is also an arrow from y back to x: 

A relation is **transitive** if whenever there are arrows from x to y and y to z ...  ...there is also an arrow from x to z: 

(If $x = z$, this means that if there are arrows from x to y and from y to x ...  ...there is also a loop from x back to x.) 

How many reflexive, irreflexive, symmetric, antisymmetric, equivalence, transitive :) relations are there on a set of n elements?

Equivalence Relations

Definition 3.19

An *equivalence relation* is a relation on a set A that is reflexive, symmetric, and transitive.

Definition 3.20

For an equivalence relation θ on a set A and for $a \in A$, the set of elements of A that are equivalent to a is called the *equivalence class* of a and is denoted as $[a]_\theta$:

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Define the relation θ on $\mathbb{Z}$, such that

$$x \,\theta\, y \iff x \equiv y \pmod 3$$

$$[0]_\theta = 3\mathbb{Z}, \qquad [1]_\theta = 3\mathbb{Z} + 1, \qquad [2]_\theta = 3\mathbb{Z} + 2$$

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"same parity" on $\{1, 2, 3, 4\}$: classes $\{1, 3\}$ and $\{2, 4\}$.

Intersections of Equivalence Relations

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The intersection of two equivalence relations (on the same set) is an equivalence relation.

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Proof sketch:

- $\mathrm{id} \subseteq \sigma$ and $\mathrm{id} \subseteq \theta \Rightarrow \mathrm{id} \subseteq \sigma \cap \theta$
- $\widehat{(\sigma \cap \theta)} = \widehat{\sigma} \cap \widehat{\theta} = \sigma \cap \theta$
- $(\sigma \cap \theta)^2 \subseteq \sigma^2 \cap \theta^2 \subseteq \sigma \cap \theta$

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What about $\sigma \cup \theta$?

Partitions and Quotient Sets

Definition 3.21

A *partition* of a set A is a set of mutually disjoint subsets of A that cover A, i.e., a set $\{S_i \mid i \in \mathcal{I}\}$ of sets S_i (for some index set $\mathcal{I}$) satisfying

$$S_i \cap S_j = \varnothing \quad \text{for } i \neq j \qquad \text{and} \qquad \bigcup_{i \in \mathcal{I}} S_i = A.$$

Definition 3.22

The set of equivalence classes of an equivalence relation θ, denoted by

$$A/\theta \stackrel{\text{def}}{=} \{[a]_\theta \mid a \in A\},$$

is called the *quotient set* of A *by* θ, or simply A *modulo* θ, or A **mod** θ.

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Example: "same parity" on $\{1, 2, 3, 4\}$:

$$A/\theta = \big\{\{1, 3\}, \{2, 4\}\big\}$$

is a partition of A. And $\mathbb{Z}/\equiv_3 = \{3\mathbb{Z},\ 3\mathbb{Z}+1,\ 3\mathbb{Z}+2\}$.

Equivalence Relations $\Leftrightarrow$ Partitions

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And the reverse direction holds too: every partition P of A induces the equivalence relation

$$x \; \theta_P \; y : \Longleftrightarrow \; x \text{ and } y \text{ lie in the same block of } P.$$

Partial Orders

Definition 3.23

A *partial order* (or simply an *order relation*) on a set A is a relation that is reflexive, antisymmetric, and transitive. A set A together with a partial order $\preceq$ on A is called a *partially ordered set* (or simply *poset*) and is denoted as $(A; \preceq)$.

Definition 3.24

For a poset $(A; \preceq)$, two elements a and b are called *comparable* if $a \preceq b$ or $b \preceq a$; otherwise they are called *incomparable*.

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In $(\mathcal{P}(\{1, 2\}); \subseteq)$ the sets $\{1\}$ and $\{2\}$ are incomparable.

Total Orders and Covering

Definition 3.25

If any two elements of a poset $(A; \preceq)$ are comparable, then A is called *totally ordered* (or *linearly ordered*) by $\preceq$.

Definition 3.26

In a poset $(A; \preceq)$ an element b is said to *cover* an element a if $a \prec b$ and there exists no c with $a \prec c$ and $c \prec b$ (i.e., between a and b).

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In $(\{1, 2, 3, 4, 6, 12\}; |)$: 4 covers 2, but 12 does *not* cover 2 because $2 \prec 4 \prec 12$.

Hasse Diagrams

Definition 3.27

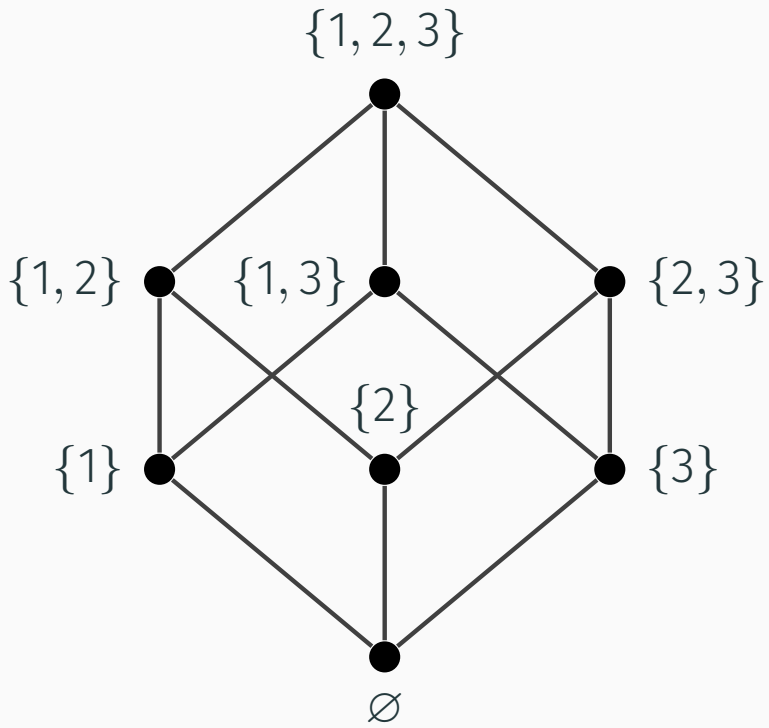
The *Hasse diagram* of a (finite) poset $(A; \preceq)$ is the directed graph whose vertices are labeled with the elements of A and where there is an edge from a to b if and only if b covers a.

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Convention: loops, arrow heads and implied (transitive) edges are left out; edges point upwards. $(\mathcal{P}(\{1,2,3\});\ \subseteq)$



Direct Product of Posets

Definition 3.28

The *direct product* of posets $(A; \preceq)$ and $(B; \sqsubseteq)$, denoted $(A; \preceq) \times (B; \sqsubseteq)$, is the set $A \times B$ with the relation $\leq$ (on $A \times B$) defined by

$$(a_1, b_1) \leq (a_2, b_2) \overset{\text{def}}{\iff} a_1 \preceq a_2 \wedge b_1 \sqsubseteq b_2.$$

Theorem 3.12

$(A; \preceq) \times (B; \sqsubseteq)$ is a partially ordered set.

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Example: $(\{0, 1\}; \leq) \times (\{0, 1\}; \leq)$ with product order from Def 3.28.

Lexicographic Order

Theorem 3.13

For given posets $(A; \preceq)$ and $(B; \sqsubseteq)$, the relation $\leq_{\text{lex}}$ defined on $A \times B$ by

$$(a_1, b_1) \leq_{\text{lex}} (a_2, b_2) \overset{\text{def}}{\iff} a_1 \prec a_2 \vee (a_1 = a_2 \wedge b_1 \sqsubseteq b_2)$$

is a partial order relation.

Lexicographic Order

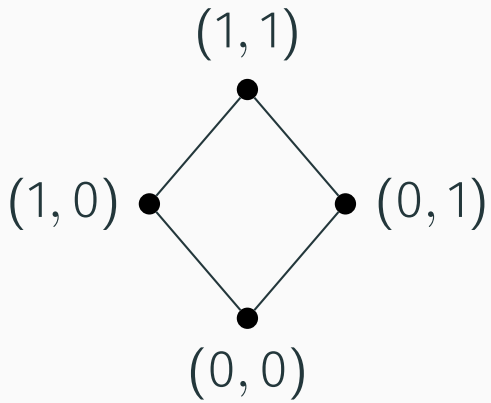
Theorem 3.13

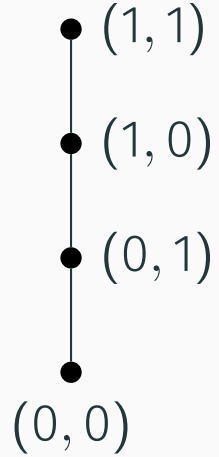
For given posets $(A; \preceq)$ and $(B; \sqsubseteq)$, the relation $\leq_{\text{lex}}$ defined on $A \times B$ by

$$(a_1, b_1) \leq_{\text{lex}} (a_2, b_2) \overset{\text{def}}{\iff} a_1 \prec a_2 \vee (a_1 = a_2 \wedge b_1 \sqsubseteq b_2)$$

is a partial order relation.

Same set $\{0, 1\} \times \{0, 1\}$: product order (left) vs. lexicographic order (right)





Extremal Elements and Bounds

Definition 3.29

Let $(A; \preceq)$ be a poset, and let $S \subseteq A$ be some subset of A. Then

1. $a \in A$ is a *minimal* (*maximal*) element of A if there exists no $b \in A$ with $b \prec a$ ($b \succ a$).
2. $a \in A$ is the *least* (*greatest*) element of A if $a \preceq b$ ($a \succeq b$) for all $b \in A$.
3. $a \in A$ is a *lower* (*upper*) *bound* of S if $a \preceq b$ ($a \succeq b$) for all $b \in S$.
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In $(\{2, 3, 4, 6, 12\}; |)$: minimal $= \{2, 3\}$, no least element;

maximal $=$ greatest $= 12$.

For $S = \{4, 6\}$: $\mathrm{ub} = \{12\}$, $\mathrm{lb} = \{2\}$, so $\mathrm{lub}(S) = 12$, $\mathrm{glb}(S) = 2$.

Well-Orders, Meet and Join

Definition 3.30

A poset $(A; \preceq)$ is *well-ordered* if it is totally ordered and if every non-empty subset of A has a least element.

Definition 3.31

Let $(A; \preceq)$ be a poset. If a and b (i.e., the set $\{a, b\} \subseteq A$) have a greatest lower bound, then it is called the *meet* of a and b, often denoted $a \wedge b$. If a and b have a least upper bound, then it is called the *join* of a and b, often denoted $a \vee b$.

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A poset $(A; \preceq)$ in which every pair of elements has a meet and a join is called a *lattice*.

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Every totally ordered finite poset is well-ordered.

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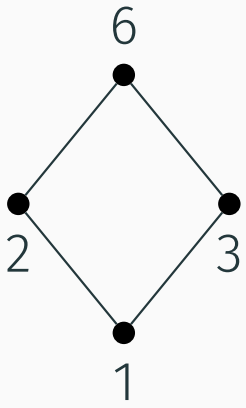
Every totally ordered finite poset is well-ordered.

$\mathbb{R}$ can also be well ordered

In $(\mathcal{P}(A); \subseteq)$: meet $= \cap$, join $= \cup$. In $(\mathbb{N}_{>0}; |)$: meet $= \gcd$, join $= \mathrm{lcm}$

Meet and Join, Visually

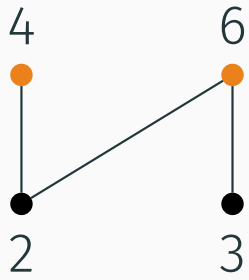
A lattice: $(\{1, 2, 3, 6\}; |)$



every pair has meet and join:

$2 \wedge 3 = 1 \qquad 2 \vee 3 = 6$

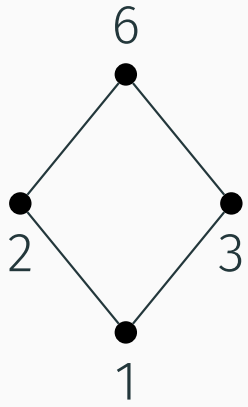
Not a lattice: $(\{2, 3, 4, 6\}; |)$



4 and 6 have *no upper bound* in this poset, so $4 \vee 6$ does not exist.

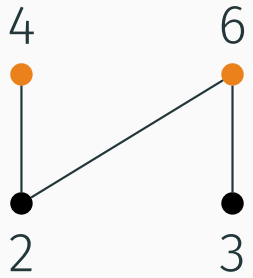
Meet and Join, Visually

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every pair has meet and join:

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4 and 6 have *no upper bound* in this poset, so $4 \vee 6$ does not exist.

$2 \wedge 3 = 1 \qquad 2 \vee 3 = 6$

A lattice fails as soon as some pair has no least upper bound

(or no greatest lower bound).

Exercises

Union of Equivalence Relations

Prove or disprove: If σ and ρ are two equivalence relations (on the same set), then $\sigma \cup \rho$ is also an equivalence relation.

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Solution

False. On $A = \{1, 2, 3\}$ *let*

$$\sigma = \mathrm{id} \cup \{(1, 2), (2, 1)\}, \qquad \rho = \mathrm{id} \cup \{(2, 3), (3, 2)\}.$$

Both are equivalence relations, but $(1, 2), (2, 3) \in \sigma \cup \rho$ *and* $(1, 3) \notin \sigma \cup \rho$*, so transitivity fails.*

Union of Partial Orders

Prove or disprove: If σ and ρ are two partial order relations (on the same set), then $\sigma \cup \rho$ is also a partial order relation.

Union of Partial Orders

Prove or disprove: If σ and ρ are two partial order relations (on the same set), then $\sigma \cup \rho$ is also a partial order relation.

Solution

False. On $A = \{1, 2\}$ *let*

$$\sigma = \{(1,1), (2,2), (1,2)\}, \qquad \rho = \{(1,1), (2,2), (2,1)\}.$$

Both are partial orders ($\leq$ *and* $\geq$)*, but* $\sigma \cup \rho$ *contains* $(1,2)$ *and* $(2,1)$ *with* $1 \neq 2$*, so antisymmetry fails.*

Classify the Relation

A relation ρ on $\mathbb{Z}$ is defined by

$$x \rho y :\iff xy \geq 0.$$

Which of the properties reflexive, irreflexive, symmetric, antisymmetric, transitive hold?

Classify the Relation

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Which of the properties reflexive, irreflexive, symmetric, antisymmetric, transitive hold?

Solution

- *Reflexive:* $x^2 \geq 0$ *for all* x.
- *Symmetric:* $xy = yx$.
- *Not transitive:* $-1 \rho 0$ *and* $0 \rho 2$, *but* $-1 \not\rho 2$.
- *Not antisymmetric:* $-1 \rho 0$ *and* $0 \rho -1$, *but* $-1 \neq 0$.
- *Not irreflexive:* $(1, 1) \in \rho$.

Transitive Closure

Compute the transitive closure of

$$\rho = \{(1,2),(2,3),(3,4),(4,2)\}$$

on $A = \{1,2,3,4\}$.

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Solution

$\rho^2 = \{(1,3),(2,4),(3,2),(4,3)\}$, $\rho^3 = \{(1,4),(2,2),(3,3),(4,4)\}$, *and* $\rho^4 = \rho$, *so*

$$\rho^* = \{(1,2),(2,3),(3,4),(4,2),\ (1,3),(2,4),(3,2),(4,3),\\ (1,4),(2,2),(3,3),(4,4)\}$$

A Relation on $\mathbb{Z} \times \mathbb{Z}$

A relation σ on $\mathbb{Z} \times \mathbb{Z}$ is defined by

$$(a, b)\ \sigma\ (c, d) :\iff a + d = b + c.$$

Prove that σ is an equivalence relation and describe its equivalence classes.

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Prove that σ is an equivalence relation and describe its equivalence classes.

Note $a + d = b + c \iff d - c = b - a$: pairs are equivalent iff they have the same *difference*. Classes are diagonals:

$$[(a, b)] = \{(x, x + (b - a)) \mid x \in \mathbb{Z}\}, \qquad (\mathbb{Z} \times \mathbb{Z})/\sigma \cong \mathbb{Z}.$$

Solution

- *Reflexive: For any* $(a, b) \in \mathbb{Z} \times \mathbb{Z}$*, since* $a + b = b + a$*, it follows that* $(a, b)\ \sigma\ (a, b)$*.*
- *Symmetric: Let* $(a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}$ *such that* $(a, b)\ \sigma\ (c, d)$*. Then* $a + d = b + c$*. By commutativity of addition,* $c + b = d + a$*, which means* $(c, d)\ \sigma\ (a, b)$*.*
- *Transitive: Let* $(a, b), (c, d), (e, f) \in \mathbb{Z} \times \mathbb{Z}$ *such that* $(a, b)\ \sigma\ (c, d)$ *and* $(c, d)\ \sigma\ (e, f)$*. Then* $a + d = b + c$ *and* $c + f = d + e$*. Adding these equations gives:*

$$(a + d) + (c + f) = (b + c) + (d + e) \implies a + f = b + e$$

Thus, $(a, b)\ \sigma\ (e, f)$*.*

Exercise 5.7

Exercise 5.7 Old Exam Question (HS18)

Let ρ and σ be two equivalence relations on a set A. Prove that if $\rho \circ \sigma = \sigma \circ \rho$ then $\rho \circ \sigma$ is an equivalence relation. That is, prove that it is reflexive, symmetric and transitive.

(a) reflexive

(b) symmetric

(c) transitive

Old Exam Task HS 24

d) ($\star$) Let $(X, \preceq)$ be a poset. Let $\prec$ be the relation on X defined as follows:

$$a \prec b \iff a \preceq b \ \wedge \ a \neq b.$$

Prove or disprove that $\prec$ is transitive.

(6 Points)

Solution

Let $a, b, c \in X$ such that $a \prec b$ and $b \prec c$. By the definition of $\prec$ we have

$$a \preceq b \land a \neq b \quad \text{and} \quad b \preceq c \land b \neq c$$

Then by transitivity of $\preceq$ we have $a \preceq c$. It remains to show that $a \neq c$. So assume for sake of contradiction that $a = c$.

Since $b \prec c$ we have that

$$b \preceq c \land b \neq c$$

With $a = c$ we can write this as

$$b \preceq a$$

Using antisymmetry of $\preceq$ gives that $a = b$, which is a contradicition to our assumption. So $a \neq b$ and we have

$$a \preceq c \land a \neq c \implies a \prec c$$

So $\prec$ is transitive.

Old Exam Task HS 25

b) ($\star$) **Prove or disprove** the following statements. For all sets $X \neq \varnothing$ and all relations ρ and τ on X:

1.) $(\rho \cap \tau)^* \subseteq \rho^* \cap \tau^*$,

2.) $(\rho \cap \tau)^* \supseteq \rho^* \cap \tau^*$,

where σ^* denotes the transitive closure of σ. *(5 Points)*

Solution

We first disprove 2.

Let $X = \{1, 2, 3\}$ *and* $\rho = \{(1, 2)\}, \tau = \{(1, 3), (3, 2)\}$. *Then we have*

$$(\rho \cap \tau)^* = (\emptyset)^* = \emptyset$$
$$\rho^* \cap \tau^* = \{(1, 2)\} \cap \{(1, 3), (1, 2), (3, 2)\} = \{(1, 2)\}$$

So $(\rho \cap \tau)^* \not\supseteq \rho^* \cap \tau^*$

Solution

Proof of 1.

First notice that we can prove the equivalent claim

$$\forall n \in \mathbb{N} \quad \bigcup_{i=1}^{n} (\rho \cap \tau)^i \subseteq \bigcup_{i=1}^{n} \rho^i \cap \bigcup_{i=1}^{n} \tau^i$$

We prove this using induction over n.

For the base case $n = 1$, we have

$$\bigcup_{i=1}^{1} (\rho \cap \tau)^i = \rho \cap \tau = \rho^1 \cap \tau^1 = \bigcup_{i=1}^{1} \rho^i \cap \bigcup_{i=1}^{1} \tau^i$$

Now assume that $\bigcup_{i=1}^{n} (\rho \cap \tau)^i \subseteq \bigcup_{i=1}^{n} \rho^i \cap \bigcup_{i=1}^{n} \tau^i$ holds.

Let $(x, y) \in \bigcup_{i=1}^{n+1} (\rho \cap \tau)^i$. We proceed by cases.

Solution

Case 1: $(x, y) \notin \bigcup_{i=1}^{n} (\rho \cap \tau)^i$. *So* $(x, y) \in (\rho \cap \tau)^{n+1}$.

Using the definition of composition, we get

$$
\begin{aligned}
& (x, y) \in \bigcup_{i=1}^{n+1} (\rho \cap \tau)^i \\
\implies & \exists z \, (x, z) \in \bigcup_{i=1}^{n} (\rho \cap \tau)^i \wedge (z, y) \in (\rho \cap \tau) \\
\implies & \exists z \, (x, z) \in \left(\bigcup_{i=1}^{n} \rho^i \cap \bigcup_{i=1}^{n} \tau^i \right) \wedge (z, y) \in (\rho \cap \tau) \\
\implies & \exists z \, (x, z) \in \bigcup_{i=1}^{n} \rho^i \wedge (x, z) \in \bigcup_{i=1}^{n} \tau^i \wedge (z, y) \in \rho \wedge (z, y) \in \tau \\
\implies & (x, y) \in \bigcup_{i=1}^{n+1} \rho^i \wedge (x, y) \in \bigcup_{i=1}^{n+1} \tau^i
\end{aligned}
$$

Solution

Case 2: $(x, y) \in \bigcup_{i=1}^{n} (\rho \cap \tau)^i$

$$
\begin{aligned}
& (x, y) \in \bigcup_{i=1}^{n+1} (\rho \cap \tau)^i \\
\implies & (x, y) \in \bigcup_{i=1}^{n} \rho^i \cap \bigcup_{i=1}^{n} \tau^i \\
\implies & (x, y) \in \bigcup_{i=1}^{n} \rho^i \wedge (x, y) \in \bigcup_{i=1}^{n} \tau^i \\
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\implies & (x, y) \in \bigcup_{i=1}^{n+1} \rho^i \cap \bigcup_{i=1}^{n+1} \tau^i
\end{aligned}
$$

Hence $\bigcup_{i=1}^{n+1} (\rho \cap \tau)^i \subseteq \bigcup_{i=1}^{n+1} \rho^i \cap \bigcup_{i=1}^{n+1} \tau^i$ *and the induction holds. Since* $\rho^* = \bigcup_{i \in \mathbb{N} \setminus \{0\}} \rho^i$*, we are done.*

Questions?

Material: `https://diegostrebel.com/teaching`

